Trustworthy AI and Robustness

BMED365: Topic List T01–T07

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Overview

1. **Trustworthy AI**
   - T01: Define trustworthy AI according to EU guidelines
2. **Robustness and Uncertainty**
   - T02: Explain the concept of robustness in ML/AI
   - T03: Describe data drift and its consequences
   - T04: Explain epistemic vs. aleatoric uncertainty
3. **Human-Machine Interaction**
   - T05: Describe human-in-the-loop (HITL) systems
   - T06: Discuss the importance of continuous monitoring
4. **Security Threats**
   - T07: Know about adversarial attacks

T01: Define trustworthy AI according to EU guidelines

EU High-Level Expert Group on AI (2019):

Definition

Trustworthy AI is AI that is lawful, ethical and robust – both technically and socially.

Seven key requirements:

1. **Human agency and oversight**
2. **Technical robustness and safety**
3. **Privacy and data governance**
4. **Transparency**
5. **Diversity, non-discrimination, fairness**
6. **Societal and environmental well-being**
7. **Accountability**

Medical relevance

High-risk AI systems in healthcare must fulfill these requirements under the EU AI Act.

T02: Explain the concept of robustness in ML/AI

Robustness = the model's ability to perform reliably under variation

A robust model:

- Provides **consistent results** on similar inputs
- **Degrades gracefully** (not catastrophically) under noise
- Generalizes well to new, unseen data

Types of robustness:

- **Noise robustness:** Tolerance for random noise
- **Distributional robustness:** Changes in data distribution
- **Adversarial robustness:** Resistance to deliberate attacks
- **Temporal robustness:** Stability over time

In medicine

A robust medical AI provides reliable predictions regardless of variation in image quality, patient population, or equipment.

T03: Describe data drift and its consequences

Data drift (distributional shift):

- Difference between **training data** and **production data**
- The model encounters data that doesn't resemble what it was trained on

Types of drift:

1. **Covariate shift:** Input distribution changes (e.g. new patient demographics)
2. **Label shift:** Class prevalence changes (e.g. pandemics)
3. **Concept drift:** The relationship between input and output changes

Examples in medicine:

- Model trained on US data used in Norway
- New MRI scanner produces different image characteristics
- COVID changed admission patterns

Consequence

The model can **fail silently** – gives predictions with high confidence that are wrong!

T04: Explain the difference between epistemic and aleatoric uncertainty

Epistemic uncertainty:

- Uncertainty due to **lack of knowledge**
- "We don't know enough yet"
- **Can be reduced** with more data
- Model is uncertain in areas with little training data

Example:

- Rare disease with few training examples
- Uncertainty because model lacks experience

Aleatoric uncertainty:

- **Inherent** random variation in data
- "The world is uncertain"
- **Cannot be reduced** with more data
- Noise in measurements, natural variation

Example:

- Two patients with identical features have different outcomes
- Uncertainty because outcome is genuinely uncertain

Practical significance

Epistemic uncertainty signals when the model should "speak up" that it is uncertain, and humans should take over.

T05: Describe human-in-the-loop (HITL) systems

Human-in-the-loop (HITL):

- Humans are **integrated** in the AI system's decision process
- AI provides recommendations, humans make final decisions

Three main variants:

1. **Human-in-the-loop:** Human involved in every decision
2. **Human-on-the-loop:** Human monitors and can override
3. **Human-out-of-the-loop:** Full autonomy (not recommended in medicine)

Advantages of HITL in medicine:

- Combines AI efficiency with human expertise
- Catches AI errors before they have consequences
- Maintains clinical responsibility, builds trust gradually

EU AI Act

High-risk AI systems **require** effective human oversight.

T06: Discuss the importance of continuous monitoring

Why continuous monitoring?

- ML models **degrade over time** (model drift)
- The world changes, data changes
- Errors in production can have serious consequences

What should be monitored?

Performance metrics:

- Accuracy, precision, recall over time
- Calibration of confidence scores
- Comparison with baseline

Data characteristics:

- Input distribution (drift detection)
- Proportion of out-of-distribution inputs
- Feature statistics

Actions when problems occur:

- **Re-training** with fresh data – update the model on newer, representative data
- **Alerting and escalation** – automatic alerts on performance drops or drift
- **Fallback** – switch to simpler model or human assessment when uncertain

T07: Know about adversarial attacks

Adversarial attacks = deliberate manipulation of AI input

- Small, seemingly invisible changes that fool the model
- Can turn a correct classification into a **completely wrong** one

Examples:

Images:

- Pixel perturbations invisible to humans
- "Panda" $\rightarrow$ "Gibbon" with 99% confidence
- Stickers that fool self-driving cars

Text (LLM):

- Prompt injection
- Jailbreaking
- Bypassing safety filters

Medical risk and defenses

Risk: An attacker could manipulate an X-ray image so AI overlooks pathology, or creates false findings.
Defenses: Adversarial training, input validation, ensemble methods, robustness testing.

Summary: Trustworthy AI and Robustness

Key points:

- **T01:** EU's 7 requirements for trustworthy AI
- **T02:** Robustness – consistent results under variation
- **T03:** Distributional shift – when production data differs from training
- **T04:** Epistemic (can be reduced) vs. aleatoric (inherent) uncertainty
- **T05:** HITL – humans in the decision loop
- **T06:** Continuous monitoring – catching drift and degradation
- **T07:** Adversarial attacks – deliberate manipulation

Key message

Trustworthy AI in medicine requires robust models, human oversight, continuous monitoring and awareness of security risks. Technology alone is not enough – organizational and regulatory measures are also required.